Dynamics

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Q1. The position of an object having mass 0.1 kg as a function of time t is given as $r = (10t^2\,\hat{i} + 5t^3\,\hat{j})$ m. At t = 1 s, which of the following statements are correct?

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A. The linear momentum $p = (2\hat{i} + 1.5\hat{j})$ kg·m/s.
B. The force acting on the object $F = (2\hat{i} + 3\hat{j})$ N.
C. The angular momentum about origin $L = 15\hat{k}$ J·s.
D. The torque about origin $\tau = 20\hat{k}$ N·m.
Choose the correct answer from the options given below:

(1) A, B and C only
(2) B, C and D only
(3) A, C and D only
(4) A, B and D only

✓ Answer: (4) A, B and D only

Solution:

$v = dr/dt = (20t, 15t^2)$; at t=1: $v = (20, 15)$.
$p = mv = 0.1(20, 15) = (2, 1.5)$ — A correct.
$a = dv/dt = (20, 30t)$; at t=1: $a = (20, 30)$.
$F = ma = 0.1(20, 30) = (2, 3)$ — B correct.
r at t=1 = (10, 5), p = (2, 1.5). $L_z = x \cdot p_y - y \cdot p_x = 10(1.5) - 5(2) = 15-10 = 5$, not 15 — C is wrong.
$\tau_z = x \cdot F_y - y \cdot F_x = 10(3) - 5(2) = 30-10 = 20$ — D correct.
So A, B, D only.

Source: NTA B.Tech — 2 April 2026, Shift 1 (official NTA release, physics section) · formulaverse.in